

EDUGENIE — PERFORMANCE TESTING

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TEST CASE 1: Q&A

Input: "Which is the largest ocean?"

Expected: "The Pacific Ocean is the largest ocean on Earth."

Result: PASSED

Time: ~5 seconds

TEST CASE 2: Explanation

Input: "Quantum Computing"

Expected: Simplified explanation for a student

Result: PASSED

Time: ~7 seconds

TEST CASE 3: Explanation (Technical Topic)

Input: "Binary Search Algorithm"

Expected: Clear, beginner-friendly explanation

Result: PASSED

Time: ~7 seconds

TEST CASE 4: Summarization

Input: 150-word paragraph on the Industrial Revolution

Expected: A concise summary retaining key points

Result: PASSED

Time: ~8 seconds

TEST CASE 5: Quiz Generation

Input: "Solar System"

Expected: 3 MCQs with 4 options each in JSON format

Result: PASSED

Time: ~10 seconds

TEST CASE 6: Learning Path

Input: "SQL"

Expected: Beginner → Intermediate → Advanced roadmap with resources

Result: PASSED

Time: ~10 seconds

TEST CASE 7: Empty Input Handling

Action: Click Submit with empty input field

Result: HTML validation blocks submission (required attribute)

Status: PASSED

TEST CASE 8: API Rate Limit

Action: Generate 25 requests rapidly

Result: 429 RESOURCE_EXHAUSTED shown in output box, no crash

Status: PASSED

TEST CASE 9: Invalid JSON from Gemini

Action: Occasionally Gemini returns extra text around JSON

Result: Regex extraction recovers clean JSON correctly

Status: PASSED

TEST CASE 10: Mobile Responsiveness

Action: Open the app on a phone in the same Wi-Fi network

Result: All 5 cards render properly and remain functional

Status: PASSED

PERFORMANCE METRICS

- Average response time: 8–10 seconds
- Fastest feature: Q&A (~5 seconds)
- Slowest feature: Quiz generation (~10 seconds)
- Browser rendering: instant
- Memory usage: under 100 MB
- Concurrent users supported: 5+ on local machine

CONCLUSION

All 10 test cases passed. EduGenie performs reliably within acceptable limits for a free-tier AI application.